A/B Testing

Data Preparation

```
library(dplyr)
library(ggplot2)
library(tidyr)
library(knitr)
library(scales)
library(lubridate)

control <- read.csv("/Users/chloeshin/Downloads/archive/control_group.csv",
    sep = ";")
test <- read.csv("/Users/chloeshin/Downloads/archive/test_group.csv",
    sep = ";")

colnames(control) <- c("Campaign", "Date", "Spend", "Impressions", "Reaches",
    "Clicks", "Searches", "Views", "Cart", "Purchases")
colnames(test) <- colnames(control)

df <- rbind(control, test)
df$Date <- dmy(df$Date)

df <- df %>%
    distinct() %>%
    drop_na()
```

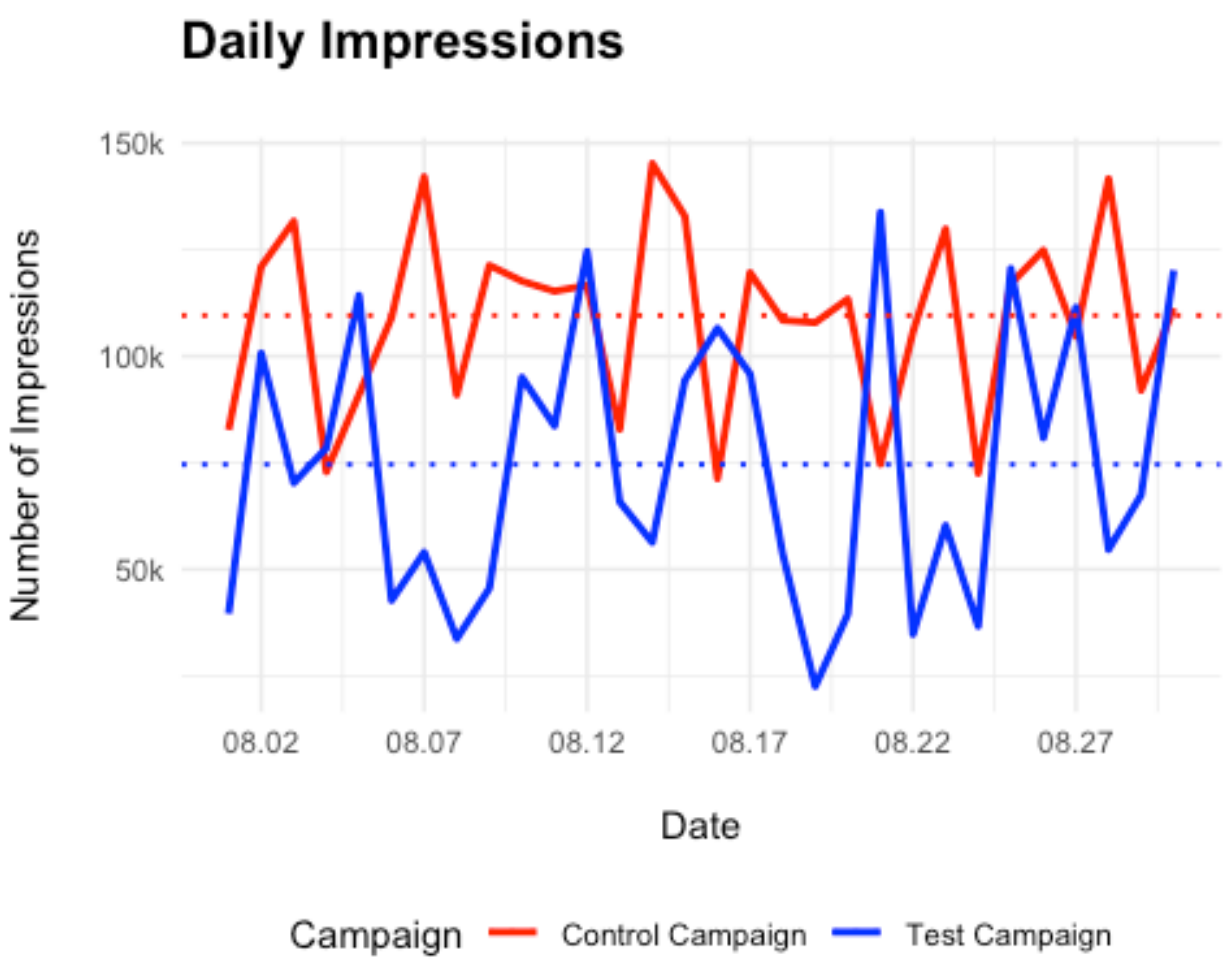
Key Metrics

```
camp_avgs <- df %>%
    group_by(Campaign) %>%
    summarise(avg_imp = mean(Impressions), avg_clicks = mean(Clicks),
        avg_purch = mean(Purchases))
```

Impressions

```
ggplot(df, aes(x = Date, y = Impressions, color = Campaign, group = Campaign)
) +
    geom_line(size = 1) + geom_hline(data = camp_avgs, aes(yintercept = avg_i
mp,
    color = Campaign), linetype = "dotted", linewidth = 0.7) + scale_y_contin
uous(labels = label_number(scale = 0.001,
    suffix = "k")) + scale_x_date(date_breaks = "5 days", date_labels = "%m.%
d") +
    scale_color_manual(values = c(`Control Campaign` = "red", `Test Campaign`
= "blue")) +
    labs(title = "Daily Impressions", x = "Date", y = "Number of Impressions"
```

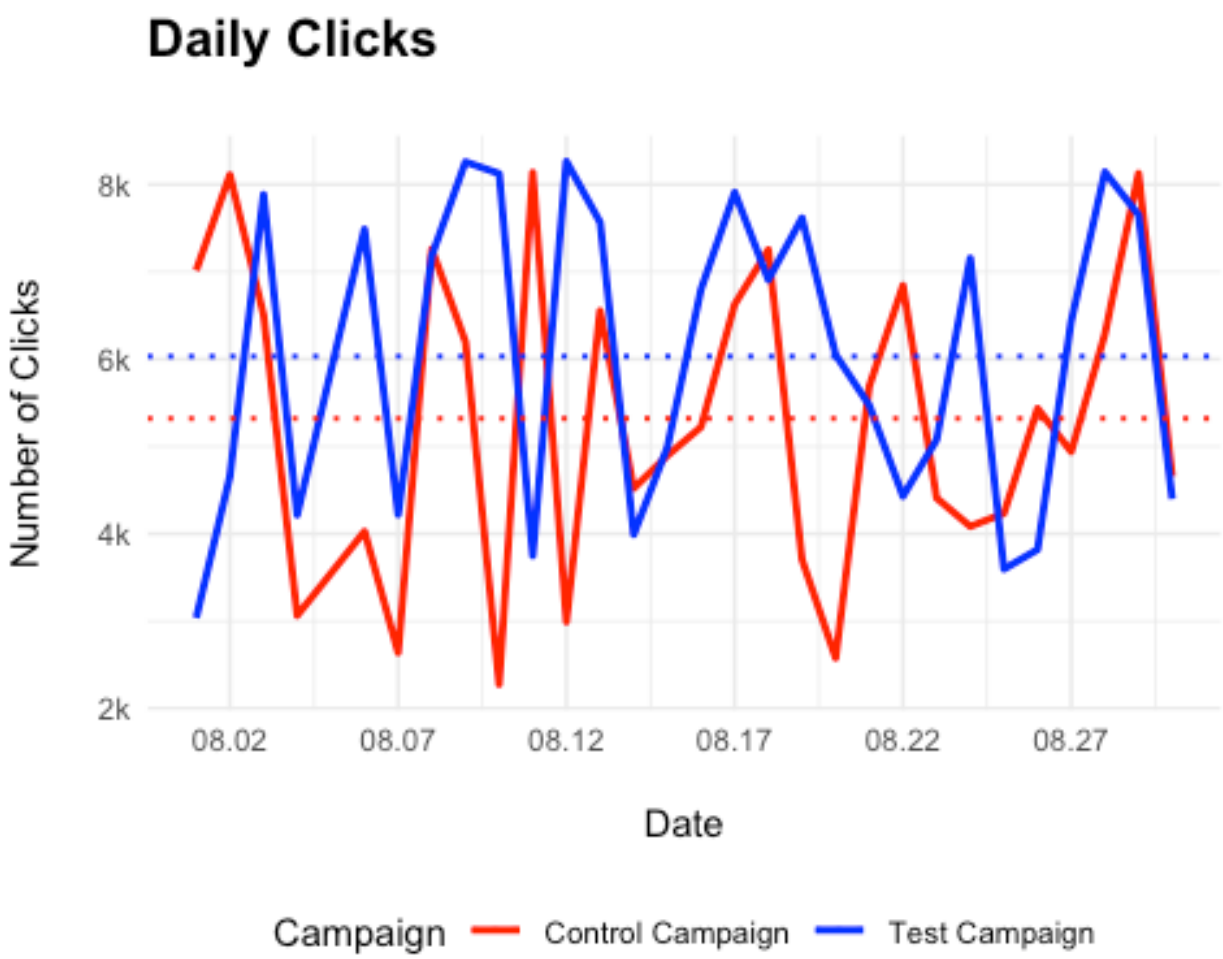
```
,
        ) + theme_minimal() + theme(plot.title = element_text(margin = margin
(b = 20),
    size = 15, face = "bold"), axis.title.x = element_text(margin = margin(t
= 15)),
    axis.title.y = element_text(margin = margin(r = 15)), legend.position = "
bottom")

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Clicks

```
ggplot(df, aes(x = Date, y = Clicks, color = Campaign, group = Campaign)) +
    geom_line(size = 1) + geom_hline(data = camp_avgs, aes(yintercept = avg_c
licks,
    color = Campaign), linetype = "dotted", linewidth = 0.7) + scale_y_contin
```

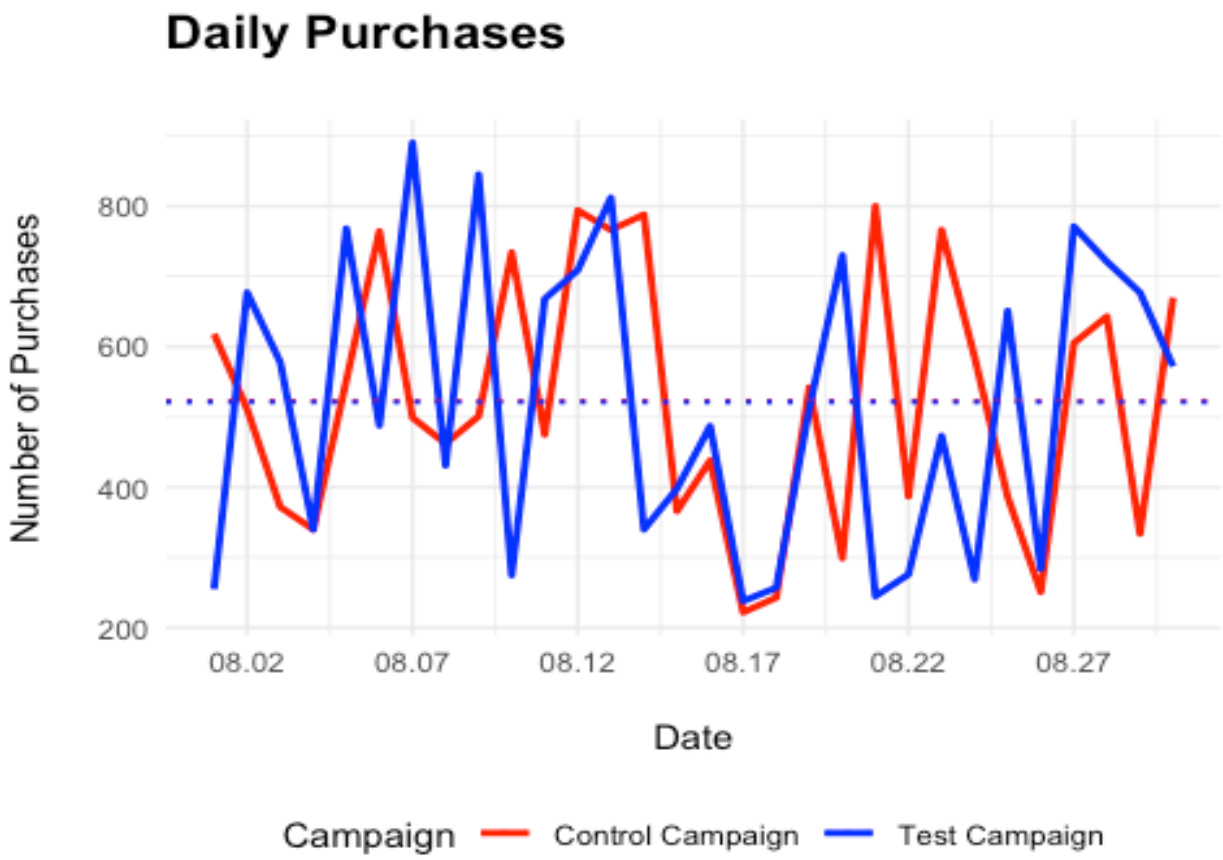
```
uous(labels = label_number(scale = 0.001,
    suffix = "k")) + scale_x_date(date_breaks = "5 days", date_labels = "%m.%
d") +
    scale_color_manual(values = c(`Control Campaign` = "red", `Test Campaign`
= "blue")) +
    labs(title = "Daily Clicks", x = "Date", y = "Number of Clicks",
        ) + theme_minimal() + theme(plot.title = element_text(margin = margin
(b = 20),
    size = 15, face = "bold"), axis.title.x = element_text(margin = margin(t
= 15)),
    axis.title.y = element_text(margin = margin(r = 15)), legend.position = "
bottom")
```



Purchases

```
ggplot(df, aes(x = Date, y = Purchases, color = Campaign, group = Campaign))
+
    geom_line(size = 1) + geom_hline(data = camp_avgs, aes(yintercept = avg_p
urch,
    color = Campaign), linetype = "dotted", linewidth = 0.7) + scale_x_date(d
```

```
ate_breaks = "5 days",
    date_labels = "%m.%d") + scale_color_manual(values = c(`Control Campaign`
= "red",
    `Test Campaign` = "blue")) + labs(title = "Daily Purchases", x = "Date",
    y = "Number of Purchases", ) + theme_minimal() + theme(plot.title = eleme
nt_text(margin = margin(b = 20),
    size = 15, face = "bold"), axis.title.x = element_text(margin = margin(t
= 15)),
    axis.title.y = element_text(margin = margin(r = 15)), legend.position = "
bottom")
```



Marketing KPIs

```
kpi_sum <- df %>%
    group_by(Campaign) %>%
    summarise(avg_CTR = round((sum(Clicks)/sum(Impressions)) * 100, digits =
2),
        avg_CVR = round((sum(Purchases)/sum(Clicks)) * 100, digits = 2),
        avg_CPC = round((sum(Spend)/sum(Clicks)), digits = 2), avg_CPP = roun
d((sum(Spend)/sum(Purchases)),
            digits = 2))
```

```r
kable(kpi_sum, col.names = c(" ", "CTR", "CVR", "CPC", "CPP"))
```

	CTR	CVR	CPC	CPP
Control Campaign	4.86	9.83	0.43	4.41
Test Campaign	8.09	8.64	0.42	4.92

Hypothesis Testing

H0: There is no statistically significant difference (KPI) between control and test campaigns.

Ha: There is a statistically significant difference (KPI) between control and test campaigns.

Click Through Rate

```r
ctr_cvr <- df %>%
    group_by(Campaign) %>%
    summarise(tot_imp = sum(Impressions), tot_clicks = sum(Clicks), tot_purch
= sum(Purchases))

ctr_test <- prop.test(ctr_cvr$tot_clicks, ctr_cvr$tot_imp)
ctr_test

##
##  2-sample test for equality of proportions with continuity correction
##
## data:  ctr_cvr$tot_clicks out of ctr_cvr$tot_imp
## X-squared = 23602, df = 1, p-value < 2.2e-16
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  -0.03274238 -0.03188490
## sample estimates:
##     prop 1     prop 2
## 0.04856521 0.08087886
```

Test Campaign has a significantly higher click through rate compared to control campaign. It is more engaging and relevant to the audience.

Conversion Rate

```r
cvr_test <- prop.test(ctr_cvr$tot_purch, ctr_cvr$tot_clicks)
cvr_test
```

```
##
##  2-sample test for equality of proportions with continuity correction
##
## data:  ctr_cvr$tot_purch out of ctr_cvr$tot_clicks
## X-squared = 140.01, df = 1, p-value < 2.2e-16
## alternative hypothesis: two.sided
## 95 percent confidence interval:
##  0.009872006 0.013824286
## sample estimates:
##     prop 1     prop 2
## 0.09825473 0.08640659
```

Test Campaign has a significantly higher conversion rate compared to control campaign. Users are more likely to purchase after landing at the test campaign.

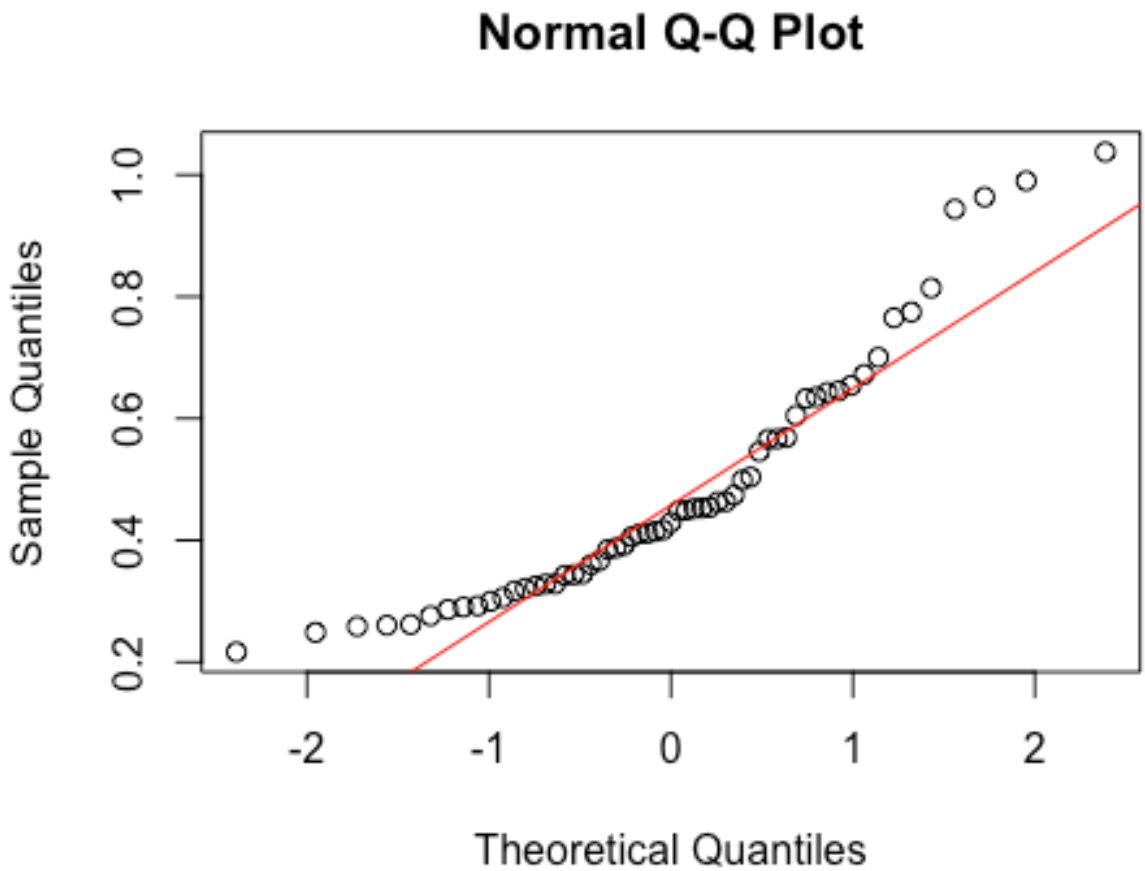
Cost Per Click

```
cpc_cpp <- df %>%
    mutate(daily_cpc = Spend/Clicks, daily_cpp = Spend/Purchases)

cpc_test <- t.test(daily_cpc ~ Campaign, data = cpc_cpp)
cpc_test
```

```
##
##  Welch Two Sample t-test
##
## data:  daily_cpc by Campaign
## t = 0.49967, df = 53.802, p-value = 0.6193
## alternative hypothesis: true difference in means between group Control Cam
paign and group Test Campaign is not equal to 0
## 95 percent confidence interval:
##  -0.07890321  0.13128327
## sample estimates:
## mean in group Control Campaign    mean in group Test Campaign
##                      0.4949079                      0.4687178
```

There is no significant difference in cost per click between two campaigns.

```
qqnorm(cpc_cpp$daily_cpc)
qqline(cpc_cpp$daily_cpc, col = "red")
```

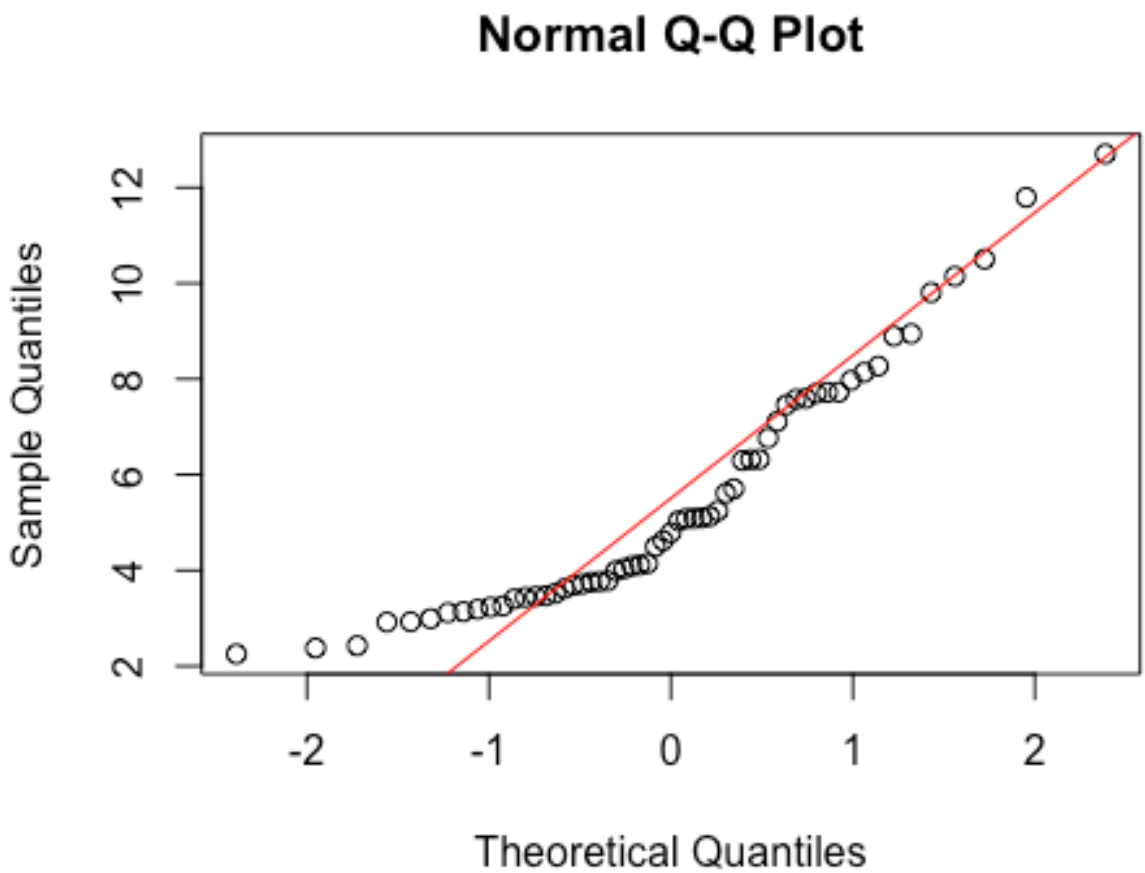


Cost Per Purchase

```
cpp_test <- t.test(daily_cpp ~ Campaign, data = cpc_cpp)
cpp_test

##
##  Welch Two Sample t-test
##
## data:  daily_cpp by Campaign
## t = -1.3136, df = 53.907, p-value = 0.1946
## alternative hypothesis: true difference in means between group Control Cam
paign and group Test Campaign is not equal to 0
## 95 percent confidence interval:
##  -2.1404415  0.4459401
## sample estimates:
## mean in group Control Campaign    mean in group Test Campaign
##                       5.052339                       5.899589
```

There is no significant difference in cost per purchase between two campaigns.

```r
qqnorm(cpc_cpp$daily_cpp)
qqline(cpc_cpp$daily_cpp, col = "red")
```



Test campaign is much efficient than the control campaign, as it has significantly higher CTR and CVR while paying the same price for each click and purchases.